INCH-POUND

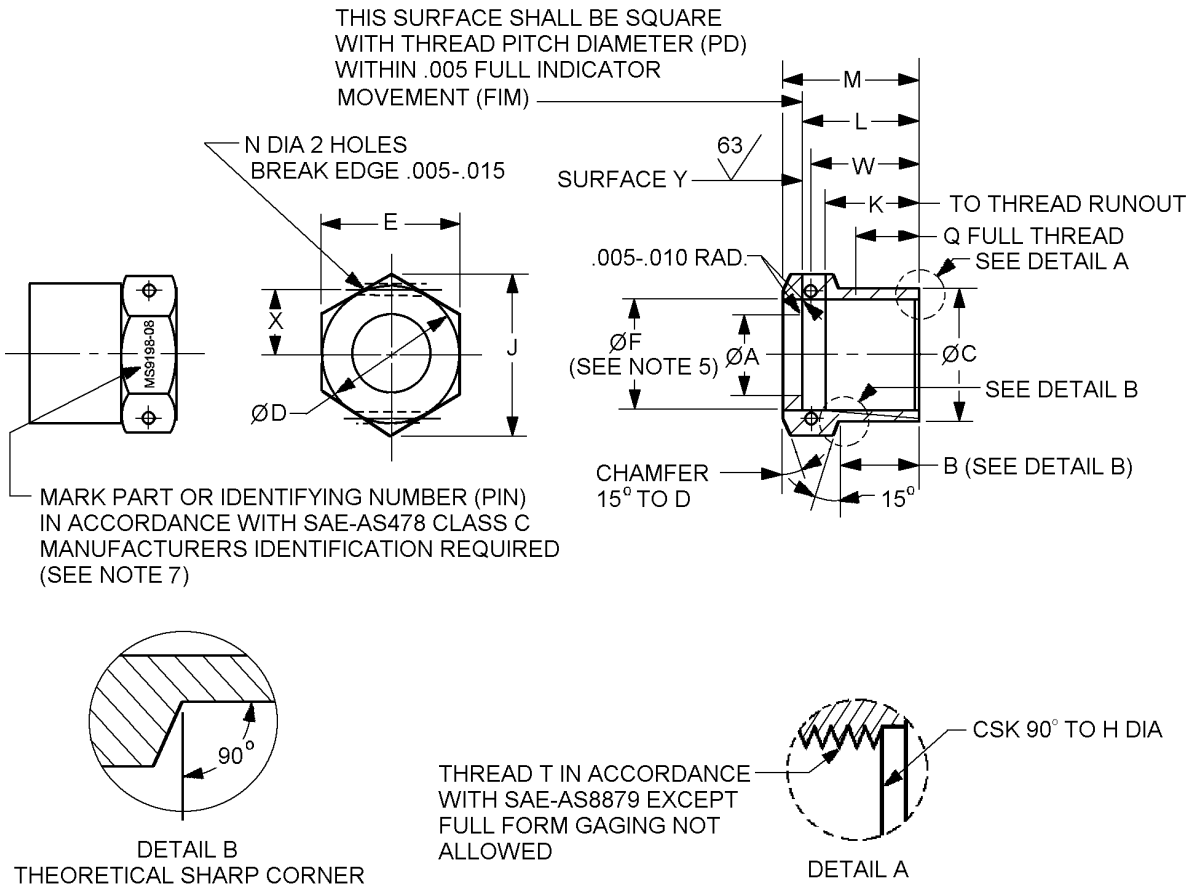
MS9198D
12 February 2010
SUPERSEDING
MS9198C
13 December 1972

DETAIL SPECIFICATION SHEET

NUT, TUBE COUPLING-CRES SAE-AMS5646

This specification is approved for use by all Departments and Agencies of the Department of Defense .

The requirements for acquiring the product described herein shall consist of this specification sheet.



Inches	mm
.005	0.13
.010	0.25
.015	0.38

FIGURE 1. Nut, tube coupling.

AMSC N/A

FSC 4730

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Dash number	Tube OD inches (mm)	Thread T SAE-AS8879	A diameter inches (mm) +.003 (0.08) -.000	B inches (mm) ±.015 (0.38)	C diameter inches (mm) ±.005 (0.13)
-02	.125 (3.18)	.3125 - 24UNJF -3B	.180 (4.57)	.435 (11.05)	.425 (10.80)
-03	.188 (4.78)	.3750 - 24UNJF -3B	.242 (6.15)	.452 (11.48)	.490 (12.45)
-04	.250 (6.35)	.4375 - 20UNJF -3B	.305 (7.75)	.467 (11.86)	.550 (13.97)
-05	.313 (7.95)	.5000 - 20UNJF -3B	.374 (9.50)	.440 (11.18)	.615 (15.62)
-06	.375 (9.53)	.5625 - 18UNJF - 3B	.440 (11.18)	.467 (11.86)	.678 (17.22)
-07	.438 (11.13)	.6250 - 18UNJF -3B	.502 (12.75)	.493 (12.52)	.740 (18.80)
-08	.500 (12.70)	.7500 - 16UNJF - 3B	.570 (14.48)	.493 (12.52)	.865 (21.97)
-09	.562 (14.27)	.8125 - 16UNJ - 3B	.634 (16.10)	.493 (12.52)	.927 (23.55)
-10	.625 (15.88)	.8750 - 14UNJF - 3B	.698 (17.73)	.617 (15.67)	.990 (25.15)
-11	.688 (17.48)	1.000 - 12UNJF - 3B	.770 (19.56)	.588 (14.94)	1.115 (28.32)
-12	.750 (19.05)	1.0625 - 12UNJ - 3B	.834 (21.18)	.588 (14.94)	1.180 (29.97)
-14	.875 (22.23)	1.1875 - 12UNJ - 3B	.961 (24.41)	.615 (15.62)	1.320 (33.53)
-16	1.000 (25.40)	1.3125 - 12UNJ - 3B	1.089 (27.66)	.652 (16.56)	1.433 (36.40)
-18	1.125 (28.58)	1.500 - 12UNJF - 3B	1.217 (30.91)	.680 (17.27)	1.620 (41.15)
-20	1.250 (31.75)	1.6250 - 12UNJ - 3B	1.347 (34.21)	.727 (18.47)	1.745 (44.32)
-24	1.500 (38.10)	1.8750 - 12UNJ - 3B	1.617 (41.07)	.880 (22.35)	1.995 (50.67)
-28	1.750 (44.45)	2.2500 - 12UNJ - 3B	1.890 (48.01)	.930 (23.62)	2.410 (61.21)
-32	2.000 (50.80)	2.500 - 12UNJ - 3B	2.167 (55.04)	1.060 (26.92)	2.660 (67.56)
-40	2.500 (63.50)	3.000 - 12UNJ - 3B	2.667 (67.74)	.860 (21.84)	3.200 (81.28)
-48	3.000 (76.20)	3.500 - 12UNJ - 3B	3.180 (80.77)	.910 (23.11)	3.700 (93.98)

FIGURE 1. Nut, tube coupling - Continued.

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Dash number	D diameter inches (mm)	E inches (mm)	F diameter inches (mm)	H inches (mm) +.030 (0.76) -.000
-02	.438 (11.13)	.430 - .439 (10.92) - (11.15)	.272 - .280 (6.91) - (7.11)	.312 (7.92)
-03	.500 (12.70)	.492 - .502 (12.50) - (12.75)	.334 - .342 (8.48) - (8.69)	.375 (9.53)
-04	.562 (14.27)	.553 - .564 (14.05) - (14.33)	.388 - .397 (9.86) - (10.08)	.438 (11.13)
-05	.625 (15.88)	.616 - .627 (15.65) - (15.93)	.451 - .459 (11.46) - (11.66)	.500 (12.70)
-06	.688 (17.48)	.679 - .690 (17.25) - (17.53)	.508 - .517 (12.90) - (13.13)	.562 (14.27)
-07	.750 (19.05)	.741 - .752 (18.82) - (19.10)	.570 - .579 (14.48) - (14.71)	.625 (15.88)
-08	.875 (22.23)	.865 - .877 (21.97) - (22.28)	.689 - .698 (17.50) - (17.73)	.750 (19.05)
-09	.937 (23.80)	.928 - .940 (23.57) - (23.88)	.751 - .760 (19.08) - (19.30)	.812 (20.62)
-10	1.000 (25.40)	.990 - 1.002 (25.15) - (25.45)	.805 - .815 (20.45) - (20.70)	.875 (22.23)
-11	1.125 (28.58)	1.114 - 1.127 (28.30) - (28.63)	.918 - .930 (23.32) - (23.62)	1.000 (25.40)
-12	1.250 (31.75)	1.239 - 1.252 (31.47) - (31.80)	.981 - .991 (24.92) - (25.17)	1.062 (26.97)
-14	1.375 (34.93)	1.364 - 1.377 (34.65) - (34.98)	1.106 - 1.116 (28.09) - (28.35)	1.188 (30.18)
-16	1.500 (38.10)	1.489 - 1.502 (37.82) - (38.15)	1.231 - 1.241 (31.27) - (31.52)	1.312 (33.32)
-18	1.750 (44.45)	1.738 - 1.752 (44.15) - (44.50)	1.419 - 1.429 (36.04) - (36.30)	1.500 (38.10)
-20	1.812 (46.02)	1.800 - 1.814 (45.72) - (46.08)	1.544 - 1.554 (39.22) - (39.47)	1.625 (41.28)
-24	2.125 (53.98)	2.113 - 2.127 (53.67) - (54.03)	1.794 - 1.804 (45.57) - (45.82)	1.875 (47.63)
-28	2.500 (63.50)	2.487 - 2.502 (63.17) - (63.55)	2.169 - 2.179 (55.09) - (55.35)	2.250 (57.15)
-32	2.750 (69.85)	2.737 - 2.752 (69.52) - (69.90)	2.419 - 2.429 (61.44) - (61.70)	2.500 (63.50)
-40	3.250 (82.55)	3.237 - 3.252 (82.22) - (82.60)	2.919 - 2.929 (74.14) - (74.40)	3.000 (76.20)
-48	3.812 (96.82)	3.798 - 3.814 (96.47) - (96.88)	3.419 - 3.429 (86.84) - (87.10)	3.500 (88.90)

FIGURE 1. Nut, tube coupling - Continued.

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Dash number	J min	K max inches (mm)	L inches (mm) ±.005 (0.13)	M inches (mm)	N inches (mm)
-02	.491 (12.47)	.425 (10.80)	.575 (14.61)	.650 (16.51)	.045 (1.14) - .050 (1.27)
-03	.561 (14.25)	.441 (11.20)	.592 (15.04)	.667 (16.94)	.057 (1.45) - .062 (1.57)
-04	.631 (16.03)	.482 (12.24)	.607 (15.42)	.682 (17.32)	
-05	.703 (17.86)	.482 (12.24)	.610 (15.49)	.690 (17.53)	.065 (1.65) - .075 (1.91)
-06	.775 (19.69)	.535 (13.59)	.637 (16.18)	.742 (18.85)	
-07	.846 (21.49)	.536 (13.61)	.663 (16.84)	.768 (19.51)	
-08	.987 (25.07)	.569 (14.45)	.723 (18.36)	.828 (21.03)	
-09	1.059 (26.90)	.569 (14.45)	.723 (18.36)	.828 (21.03)	
-10	1.130 (28.70)	.667 (16.94)	.817 (20.75)	.927 (23.55)	
-11	1.271 (32.28)	.738 (18.75)	.868 (22.05)	.968 (24.59)	
-12	1.414 (35.92)	.738 (18.75)	.868 (22.05)	.968 (24.59)	
-14	1.556 (39.52)	.765 (19.43)	.915 (23.24)	1.015 (25.78)	
-16	1.699 (43.15)	.786 (19.96)	.962 (24.43)	1.062 (26.97)	
-18	1.985 (50.42)	.800 (20.32)	.990 (25.15)	1.120 (28.45)	
-20	2.054 (52.17)	.810 (20.57)	1.017 (25.83)	1.167 (29.64)	
-24	2.411 (61.24)	.937 (23.80)	1.170 (29.72)	1.320 (33.53)	
-28	2.838 (72.09)	.945 (24.00)	1.170 (29.72)	1.370 (34.80)	
-32	3.123 (79.32)	1.050 (26.67)	1.300 (33.02)	1.500 (38.10)	
-40	3.693 (93.80)	.850 (21.59)	1.100 (27.94)	1.300 (33.02)	
-48	4.334 (110.08)	.900 (22.86)	1.150 (29.21)	1.350 (34.29)	

FIGURE 1. Nut, tube coupling - Continued.

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Dash number	Q min	W Max inches (mm)	X inches (mm) ±.005 (0.13)
-02	.300 (7.62)	.535 (13.59)	.180 (4.57) - .190 (4.83)
-03	.316 (8.03)	.552 (14.02)	.210 (5.33) - .220 (5.59)
-04	.332 (8.43)	.567 (14.40)	.245 (6.22) - .255 (6.48)
-05	.332 (8.43)	.565 (14.35)	.280 (7.11) - .290 (7.37)
-06	.367 (9.32)	.592 (15.04)	.310 (7.87) - .320 (8.13)
-07	.368 (9.35)	.618 (15.70)	.336 (8.53) - .356 (9.04)
-08	.383 (9.73)	.648 (16.46)	.400 (10.16) - .420 (10.67)
-09	.383 (9.73)	.668 (16.97)	.427 (10.85) - .447 (11.35)
-10	.453 (11.51)	.777 (19.74)	.465 (11.81) - .485 (12.32)
-11	.488 (12.40)	.818 (20.78)	.520 (13.21) - .540 (13.72)
-12	.488 (12.40)	.818 (20.78)	.580 (14.73) - .600 (15.24)
-14	.515 (13.08)	.865 (21.97)	.645 (16.38) - .665 (16.89)
-16	.536 (13.61)	.912 (23.16)	.715 (18.16) - .735 (18.67)
-18	.550 (13.97)	.945 (24.00)	.840 (21.34) - .860 (21.84)
-20	.560 (14.22)	.967 (24.56)	.890 (22.61) - .910 (23.11)
-24	.687 (17.45)	1.100 (27.94)	1.040 (26.42) - 1.060 (26.92)
-28	.695 (17.65)	1.170 (29.72)	1.255 (31.88) - 1.275 (32.39)
-32	.800 (20.32)	1.300 (33.02)	1.395 (35.43) - 1.415 (35.94)
-40	.600 (15.24)	1.100 (27.94)	1.680 (42.67) - 1.700 (43.18)
-48	.650 (16.51)	1.150 (29.21)	2.000 (50.80) - 2.020 (51.31)

FIGURE 1. Nut, tube coupling - Continued.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Tolerances linear dimensions ±.010 inch (0.25 mm) angular dimensions ±5°.
4. Surface texture unless otherwise specified shall be 125µ inches (3.17µm) except hex, reference ASME B46.1.
5. Diameters A and F shall be concentric to thread PD within .004 inch (0.010 mm) FIM. Diameter C and hexagon E shall be concentric to thread PD within .010 inch (0.25 mm) FIM.
6. Break Sharp edges .003 - .015 inch (0.08 - 0.38 mm) unless otherwise specified. Fillets .005 - .020 inch (0.13 - 0.51 mm) RAD or chamfer.
7. Multiple marking permissible provided positive identification is shown. Parts shall be marked before threading.
8. For design features purposes, this standard takes precedence over documents referenced herein.

FIGURE 1. Nut, tube coupling - Continued.

REQUIREMENTS:

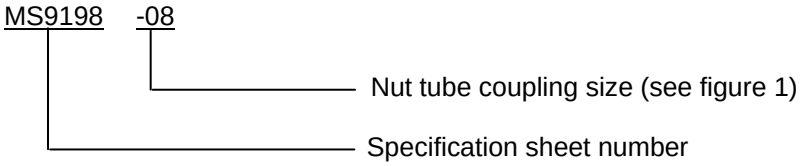
Dimensions and configurations: See figure 1.

Material: Steel, corrosion and heat-resistant in accordance with SAE-AMS5646.

Parts shall be fluorescent penetrate inspected in accordance with ASTM E1417.

Cleaning: Parts shall be degreased and immersed for not less than 20 minutes in a solution of 1 volume of nitric acid (specific gravity = 1.42) and 9 volumes of water at room temperature.

PIN:



PIN Example:

MS9198-08 identifies a nut tube coupling .500 inch (0.13 mm) tube.

Do not use unassigned PIN's.

The approximate weight for each fitting is shown in table I.

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TABLE I. Approximate weight.

Dash number	Approximate weight lb/100 (kg/100)		Dash number	Approximate weight lb/100 (kg/100)
-02	1.210 (0.55)		-11	7.859 (3.56)
-03	1.475 (0.67)		-12	9.907 (4.49)
-04	1.830 (0.83)		-14	12.164 (5.52)
-05	2.600 (1.18)		-16	13.637 (6.19)
-06	2.920 (1.32)		-18	19.830 (8.99)
-07	3.422 (1.55)		-20	20.016 (9.08)
-08	4.680 (2.12)		-24	27.093 (12.29)
-09	5.314 (2.41)		-28	41.710 (18.92)
-10	6.310 (2.86)		-32	49.000 (22.23)
			-40	60.621 (27.50)
			-48	78.529 (35.62)

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. This document references the following:

ASME B46.1
ASTM E1417
SAE-AS478
SAE-AS8879
SAE-AMS5646

CONCLUDING MATERIAL

Custodians:
Army - MI
Navy - AS
Air Force - 99
DLA - CC

Review activities:
Navy - MC
Air Force - 71

Preparing activity:
DLA - CC

(Project 4730-2009-084)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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